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UNIVERSITY COLLEGE
MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

Histology of Suprarenal gland and Urinary system

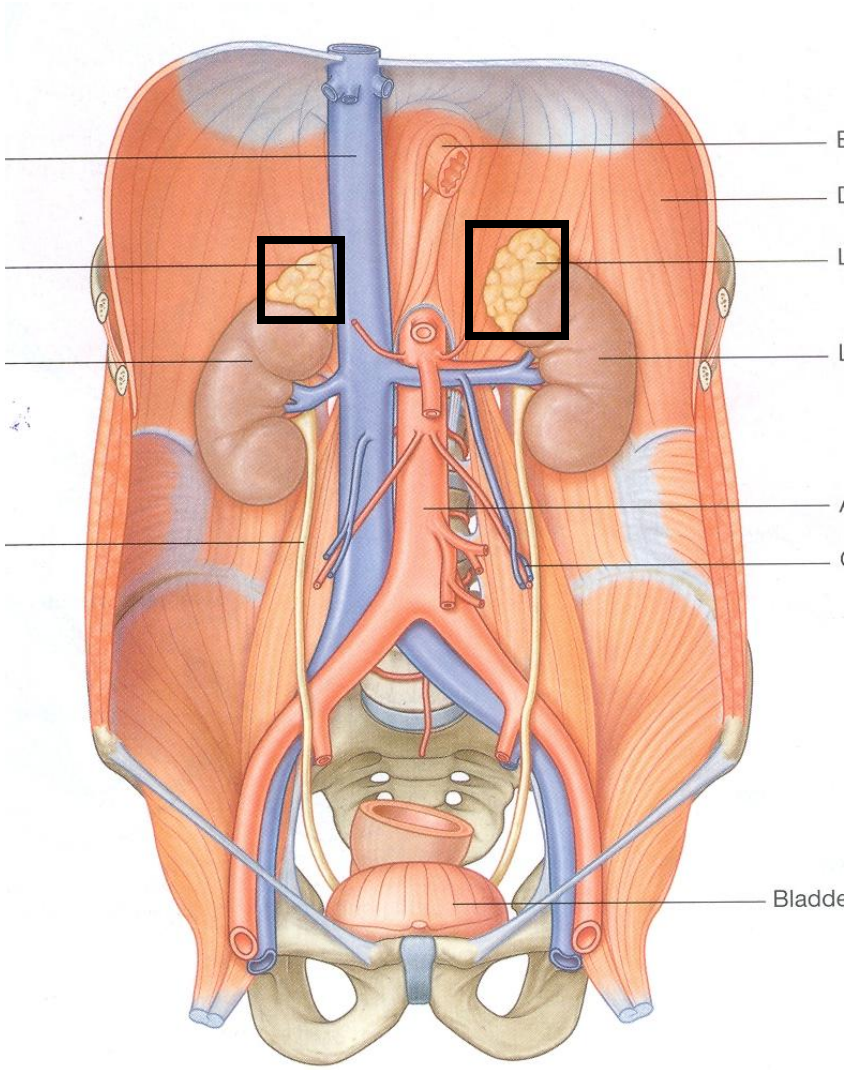


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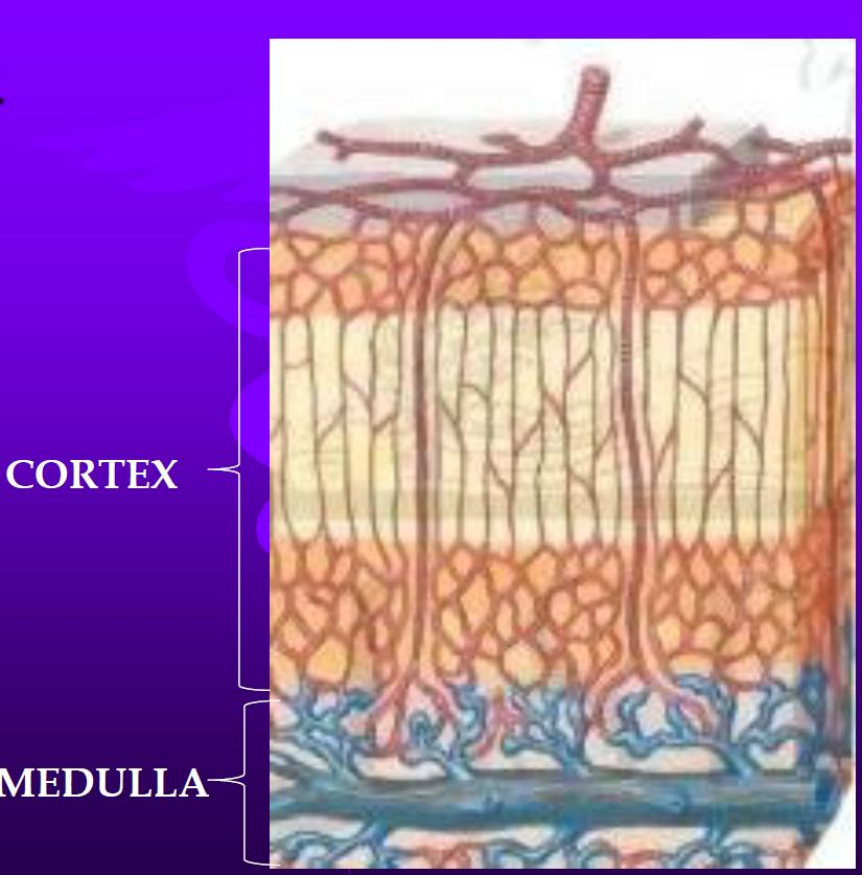
#inspire

- A pair of endocrine glands
- Situation:-on the posterior abdominal wall over the upper pole of the kidneys behind the peritoneum
- Shape:-right suprarenal gland – triangular

  left suprarenal gland -semilunar

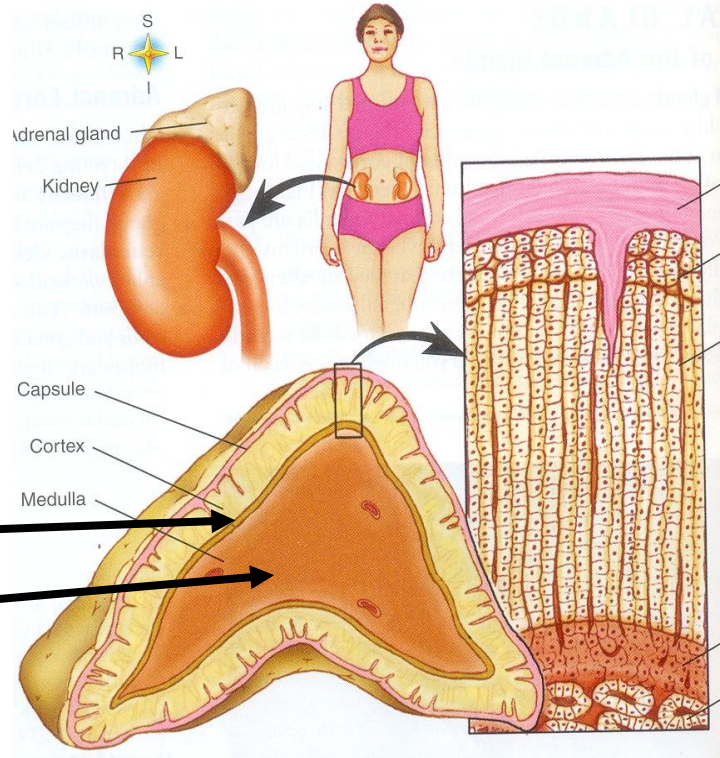


Histology



- **Covered by connective tissue capsule**
- **Septa from capsule extend into the gland substance**
- **Parts:-superficial-cortex (9/10)**

  deeper-medulla (1/10)



- # Cortex

➢ **Made up of cells arranged in cards.**

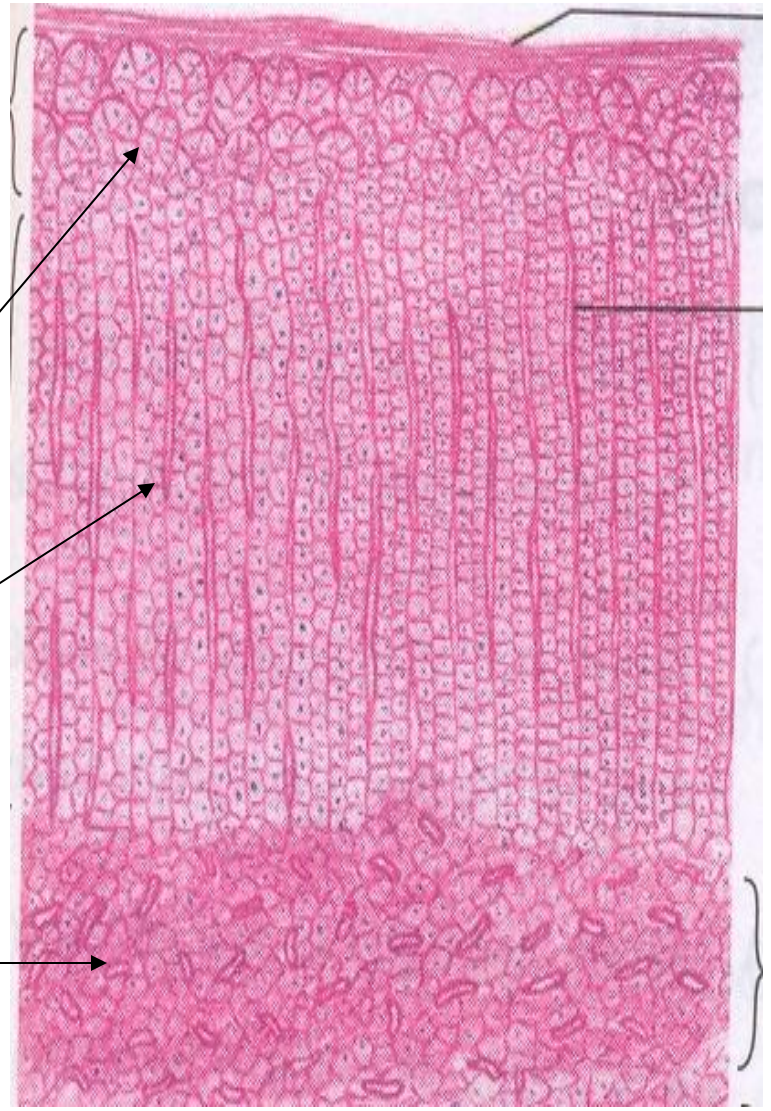
➢ **Sinusoids intervene between the cords.**

➢ **Layers of cortex**

a) **Outer most layer – zona glomerulosa (1/5th of the cortex)**

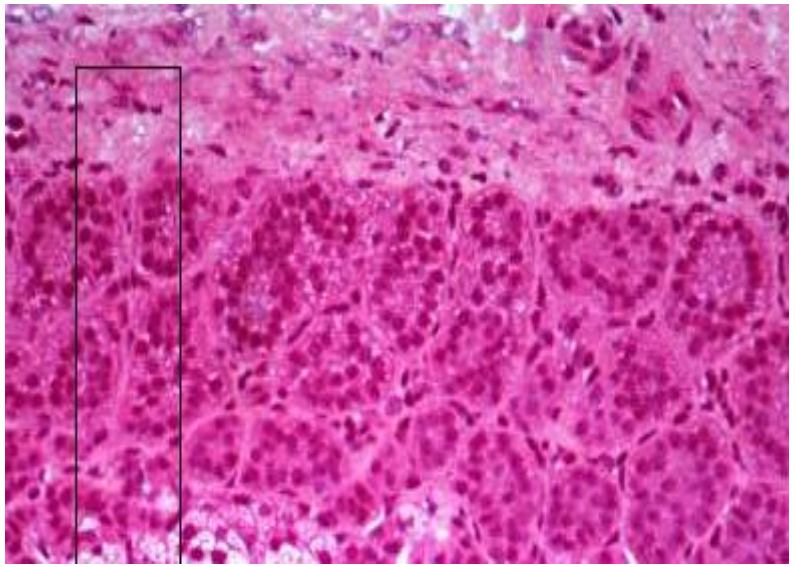
b) **Middle layer – zona fasciculata (3/5th of the cortex)**

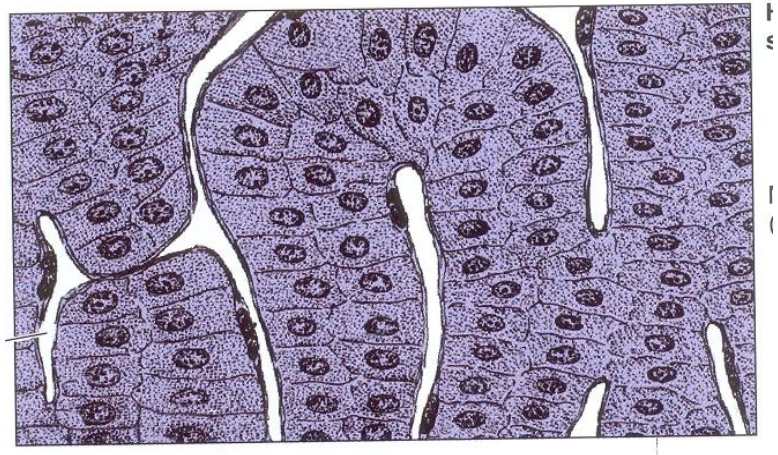
c) **Innermost layer – zona reticularis (1/5th Of the cortex)**



Zona glomerulosa

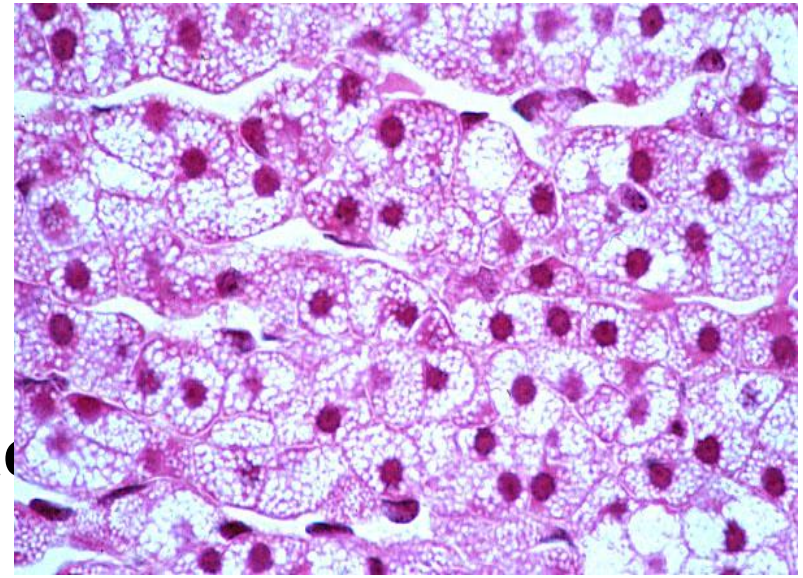
- Cells are arranged as inverted u-shaped formations
- Cells are polyhedral or columnar. Basophilic cytoplasm and deeply stained nuclei.
- Hormones – mineralocorticoids- aldosterone, deoxycorticosterone.

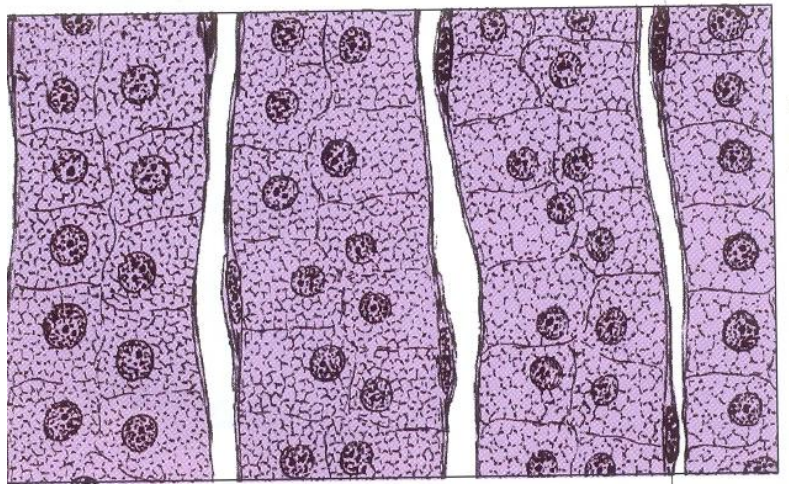




Zona fasciculata

- Cells are arranged in straight columns in two cell thickness.
- Sinusoids intervenes between the columns.
- Cells are large polyhedral basophilic cytoplasm and vasicular nuclei.
- Hormones – glucocorticoids- cortizone, cortisol dehydroepiandrosterone(DHA)





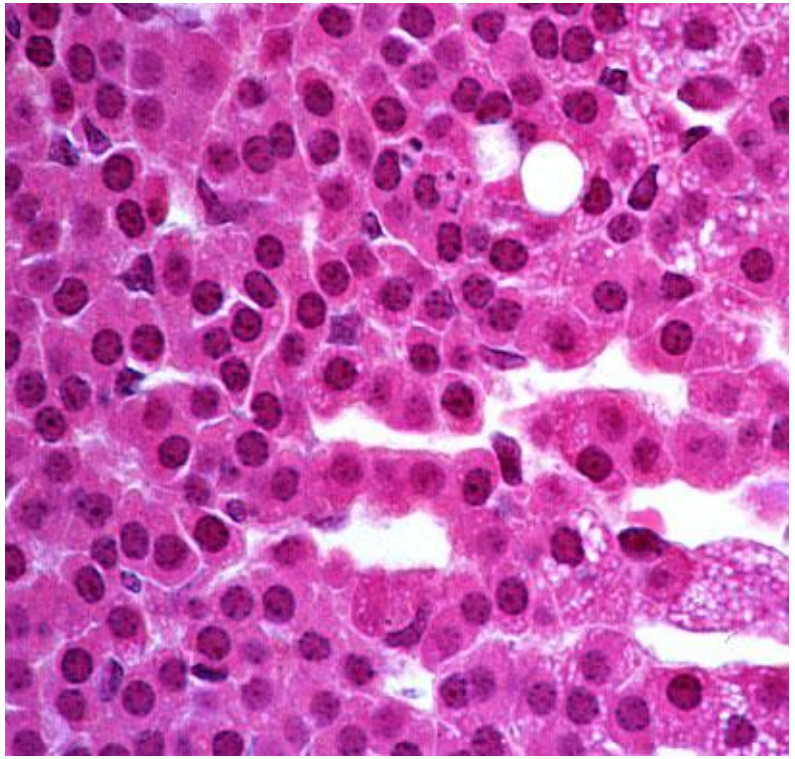
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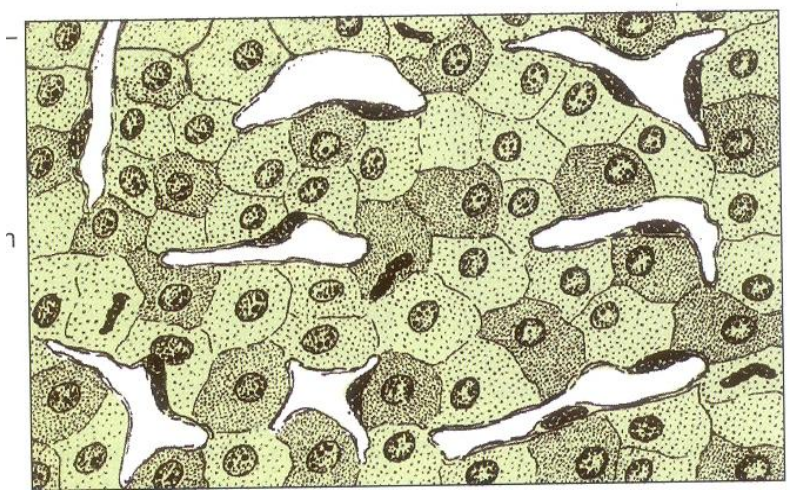
- Zona reticularis-

➢**Cords will branch and anastomose with each other to form a kind of reticulum.**

➢**Cells are large polyhedral eosinophilic cytoplasm.**

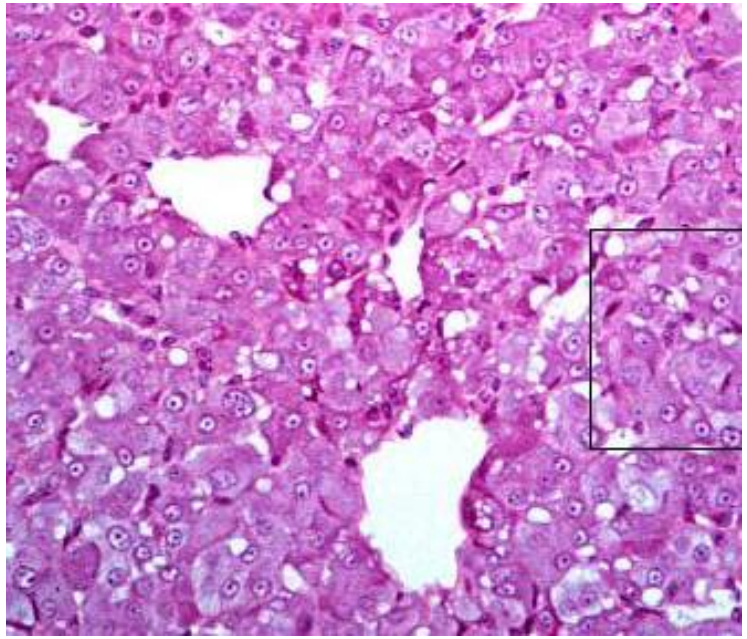
➢**Hormones – glucocorticoids sex hormones-eastrogens and androgens.**

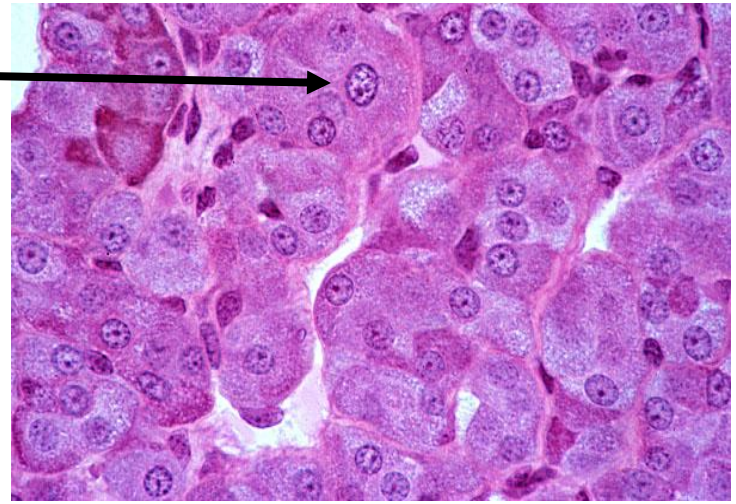




Medulla

- Distinct from the cortex both functionally and embryologically
- Made up of groups of cells seperated by wide sinusoids.
- Cells are columnar or polyhedral with basophilic cytoplasm (chromaffin cells).
- Functionally these are modified post ganglionic sympathetic neurons.
- Hormones – noradrenalin and adrenalin.





	CORTEX			MEDULLA
	ZONA GLOMERULOSA	ZONA FASCICULATA	ZONA RETICULARIS	
CELLS	Columnar, inverted u-shaped, basophilic	Polyhedral, straight columns, basophilic, lipid - spongiocytes	Polyhedral, cords branch & anastomose, eosinophilic	Groups, columnar/polyhedral, basophilic
SINUSOIDS	+	+	+	+
CELLS STAINING	Basophilic	Basophilic	Eosinophilic	Basophilic
NUCLEI	Deep staining	Vesicular	Deep staining	Deep staining
HORMONES	Mineralocorticoids	Glucocorticoids	Sex hormones	Adrenalin & Nor-adrenalin
SPECIAL FEATURES		Lipids - spongiocytes		Modified post-ganglionic sympathetic neurons

	Zone	Hormone	Functions	Control of secretion
Cortex	**Zona glomerulosa**	Mineralocorticoid (aldosterone)	It controls the water and electrolyte balance of body. It acts on the distal convoluted tubules for resorption of sodium.	Regulated by renin-angiotensin system and pituitary gland.
	Zona fasciculata	Glucocorticoids (cortisol)	It regulates carbohydrates, protein, amino acids and fat metabolism. It is also important in body's response to stress and in reducing inflammatory, allergic and immune response.	It is under the control of pituitary gland (ACTH).
	Zona reticularis	Androgen (dehydroepian-drosterone) Estrogen	These sex hormones are of little physiological significance.	It is also under the control of pituitary gland.
Medulla		Catecholamines (epinephrine and norepinephrine)	The release of these hormones prepares the body to combat emotional stress.	The secretion of adrenal medulla is in response to stimulation by preganglionic fibers from splanchnic nerves.

summary



cortex

medulla

Zona glomerulosa

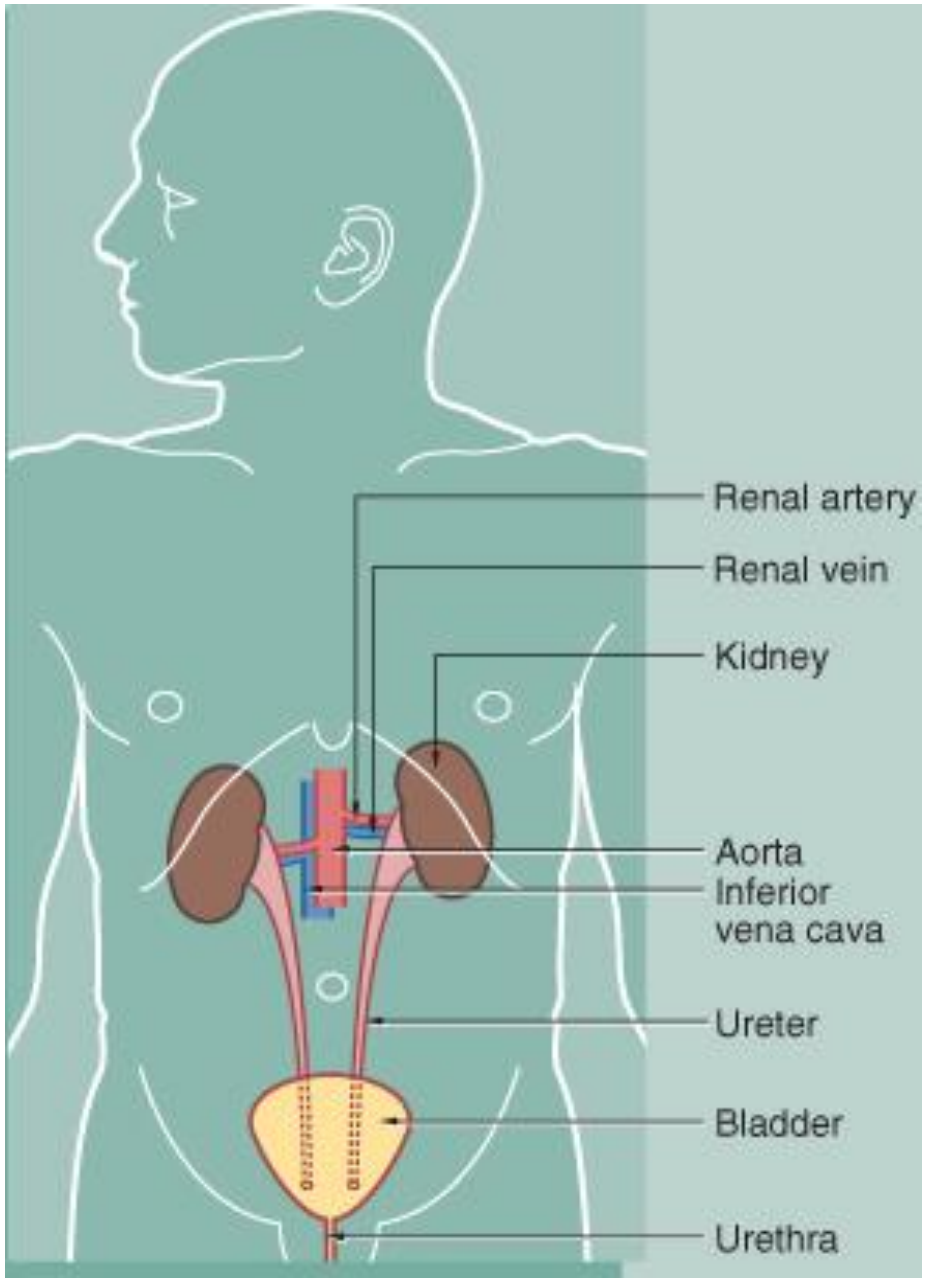
Zona fasciculata

Zona reticularis

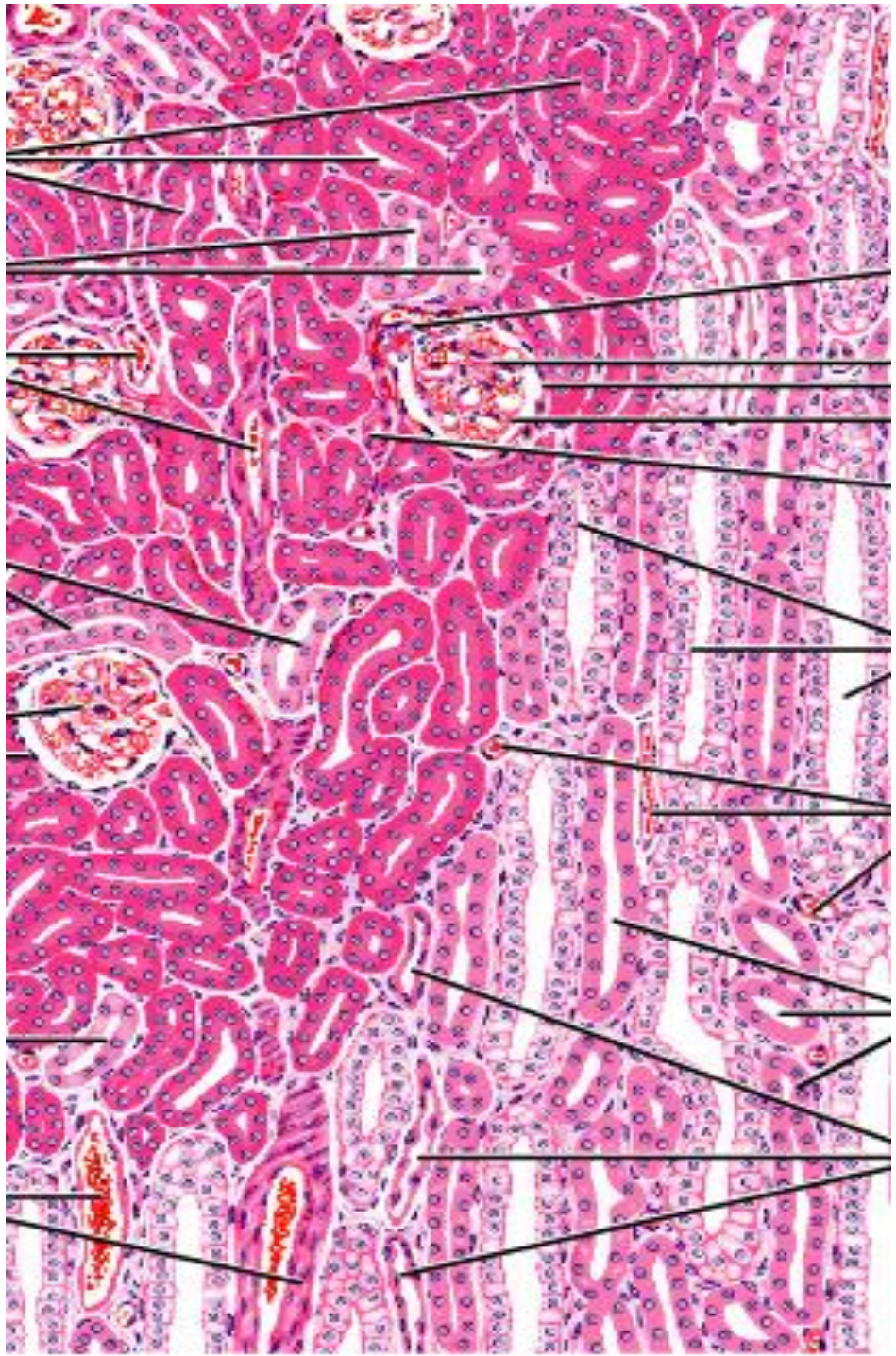
Chromaffin cells

Urinary system

- Comprises
  - Two kidneys
  - Two ureters
  - A urinary bladder
  - A urethra

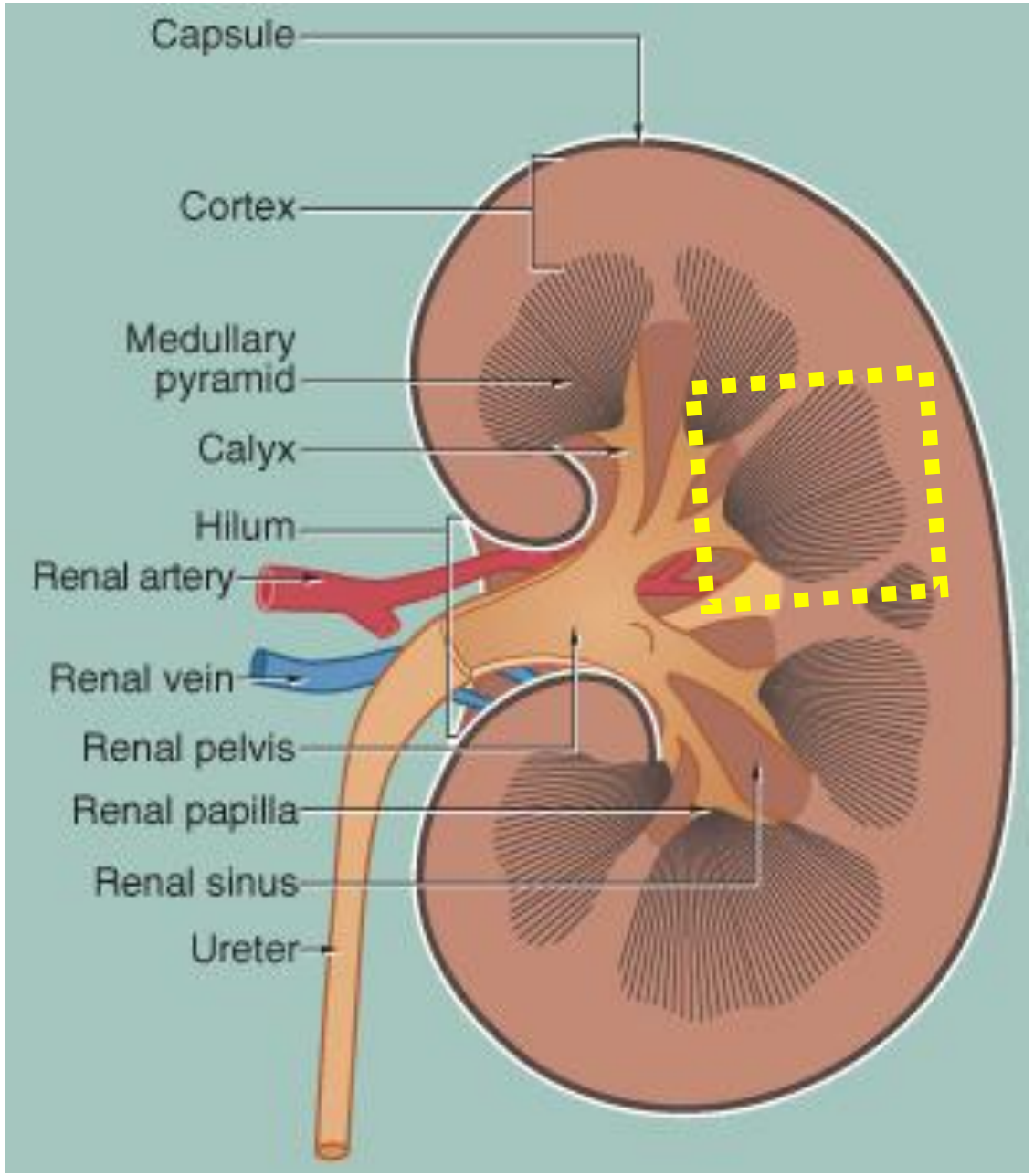


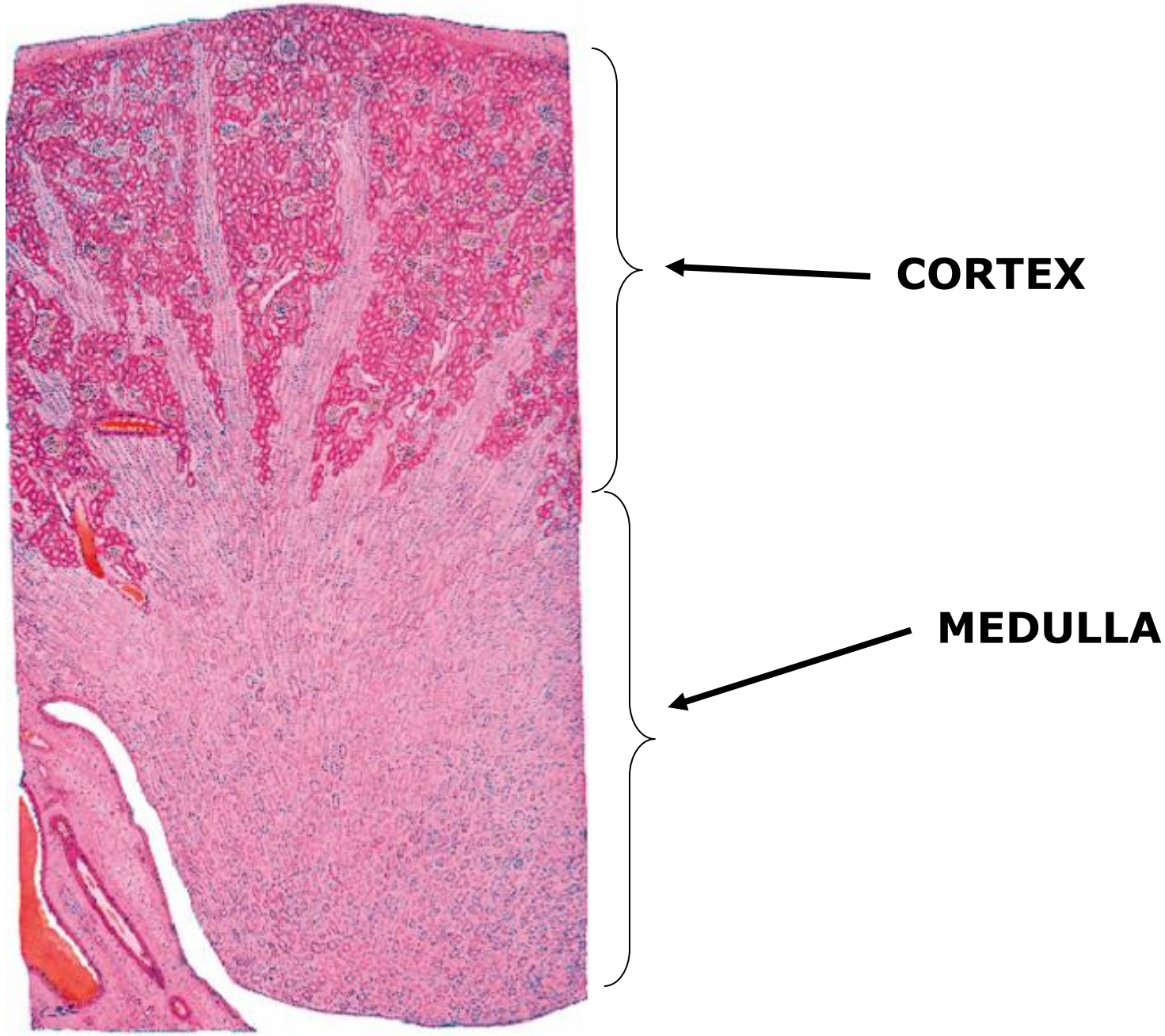
KIDNEY

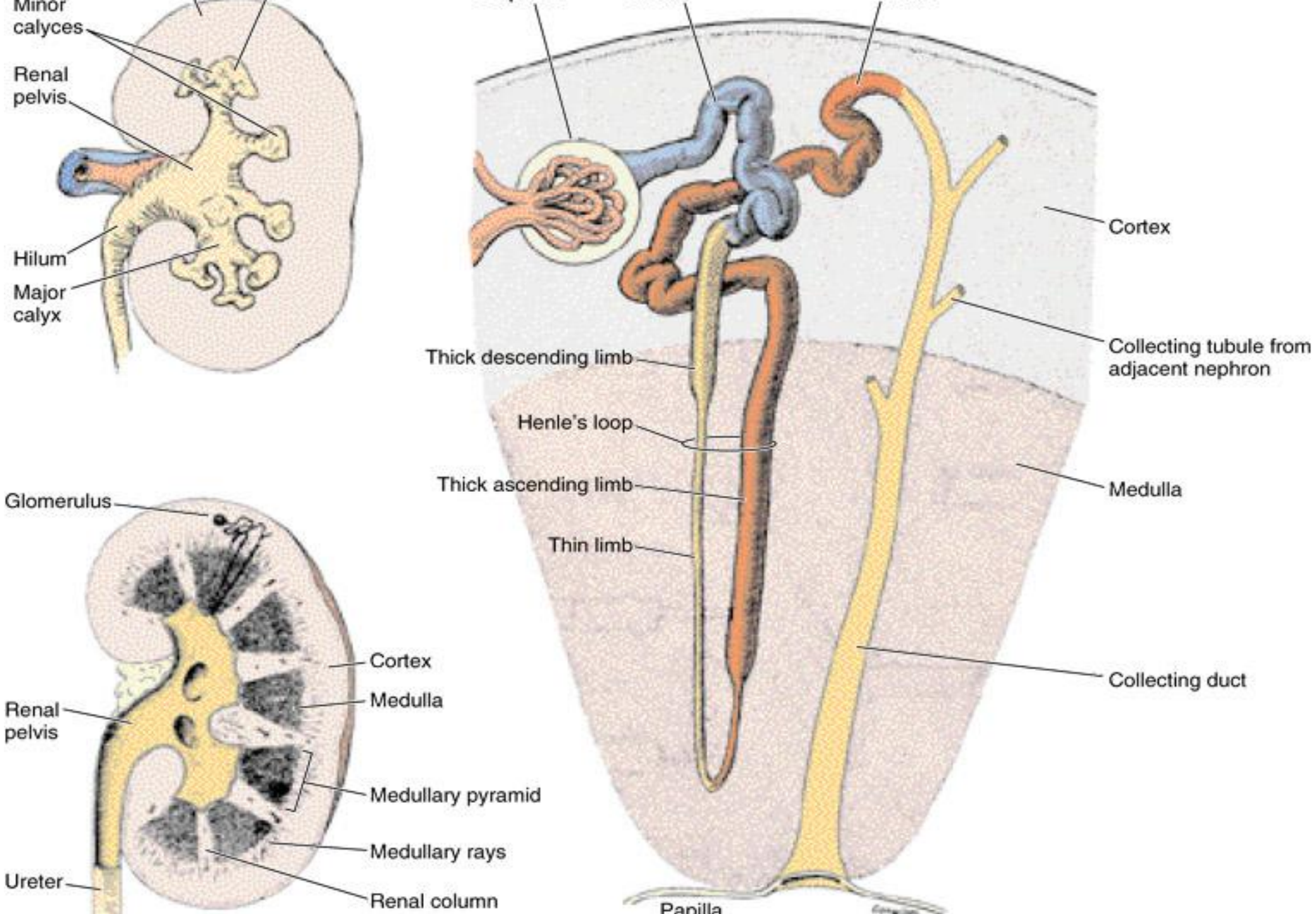


Microscopic structure of kidney

- Covered by dense capsule
- Substance is divided into: outer cortex and inner medulla
- Medulla is arranged into units – renal pyramids
- Medullary pyramids are separated by extensions of cortical tissue, called renal columns
- Functional unit of the kidney – uriniferous tubule
- Each tubule consist of nephron and collecting duct
- Each nephron consist of renal corpuscle and renal tubule

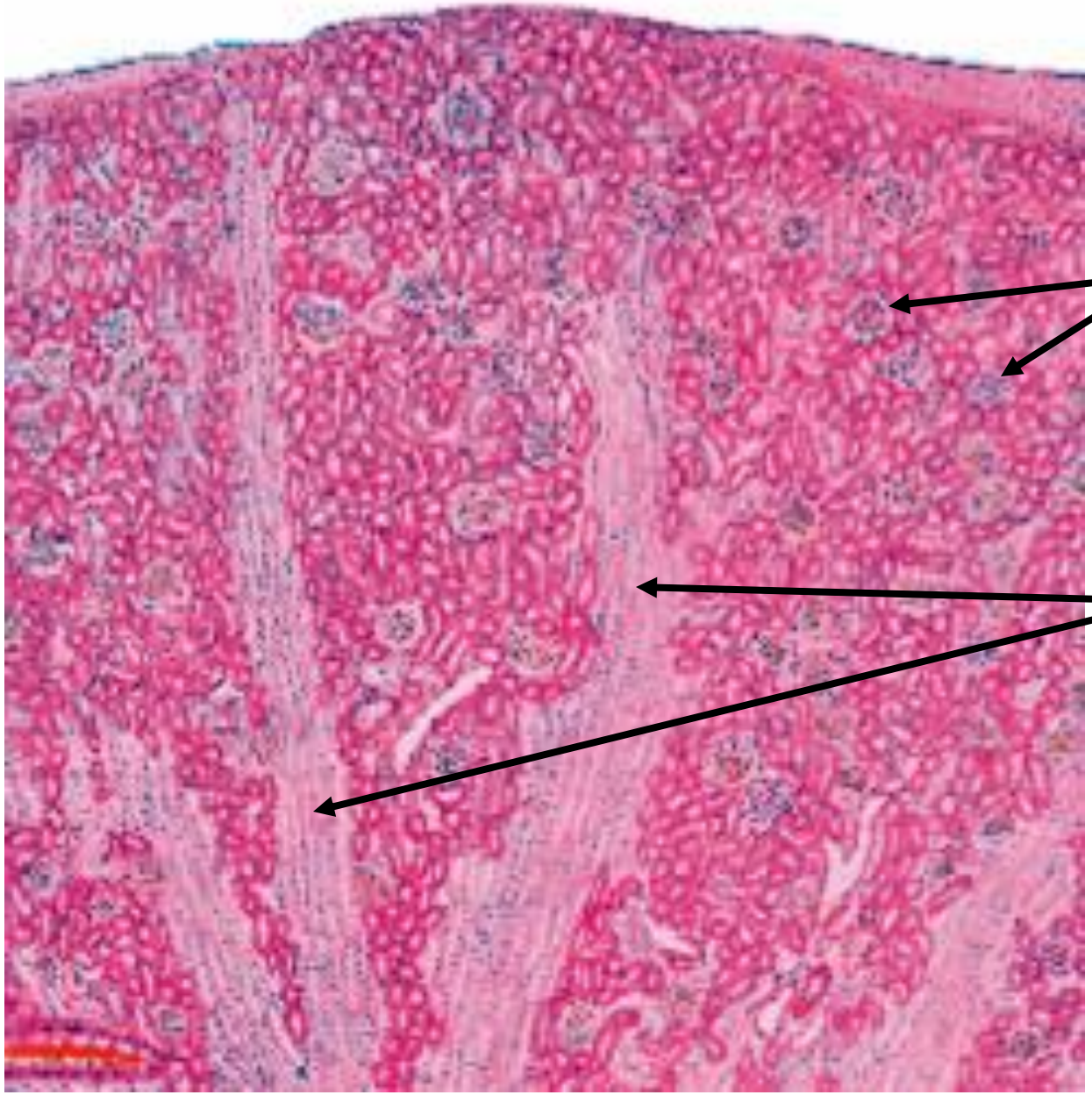

Capsule
Cortex
Medullary pyramid
Calyx
Hilum
Renal artery
Renal vein
Renal pelvis
Renal papilla
Renal sinus
Ureter


CORTEX
MEDULLA


Minor calyces
Renal pelvis
Hilum
Major calyx
Glomerulus
Cortex
Medulla
Renal pelvis
Medullary pyramid
Medullary rays
Ureter
Renal column
Thick descending limb
Henle's loop
Thick ascending limb
Thin limb
Cortex
Collecting tubule from adjacent nephron
Medulla
Collecting duct
Papilla

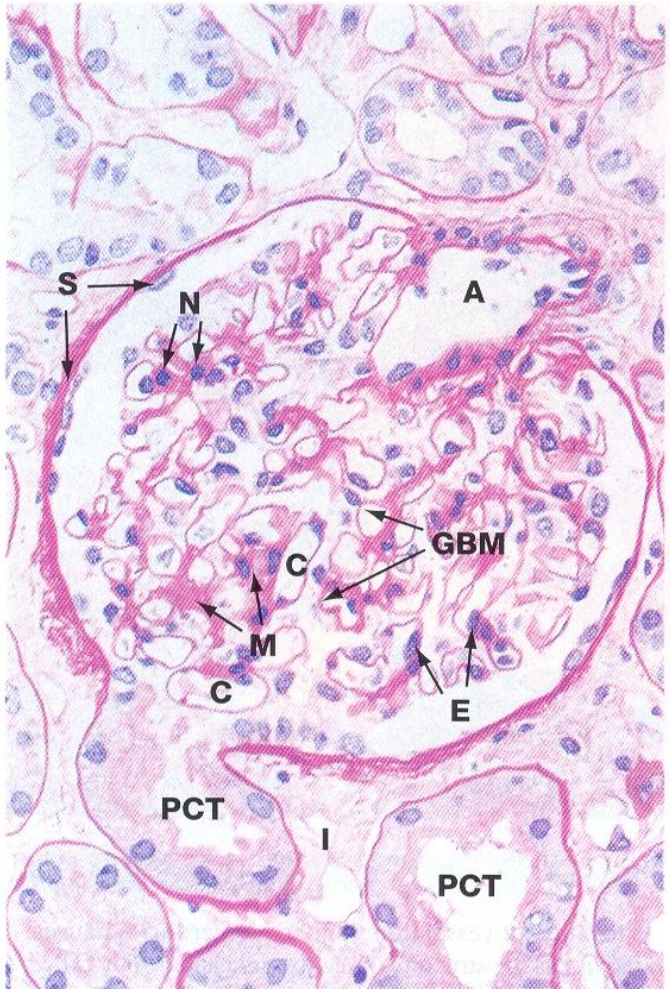
CORTEX:

- Darker outer portion
- Contains renal corpuscles, proximal and distal convoluted tubles and medullary rays
- Also contains interlobular arteries and veins
- Pale stained medullary rays formed by straight portions of proximal and distal convoluted tubules and collecting tubules

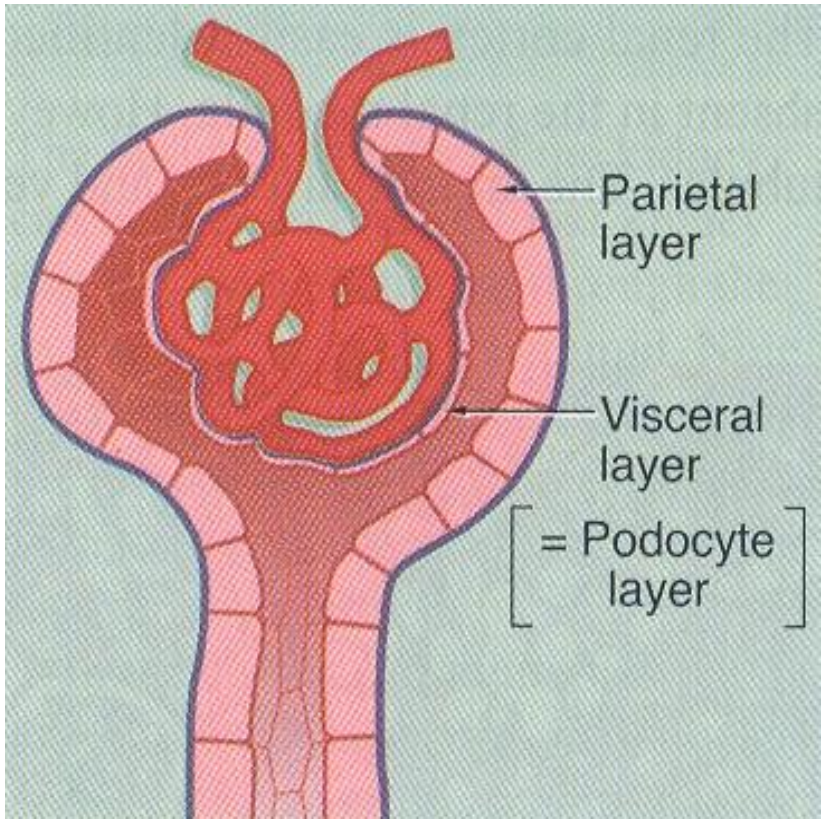

GLOMERULI
MEDULLARY RAYS

Renal corpuscle:

- Dense rounded structures
- Responsible for the filtration of plasma
- Comprises:
  - Glomerulus is formed of a tight anastomosing network of capillaries
  - Bowman's capsule



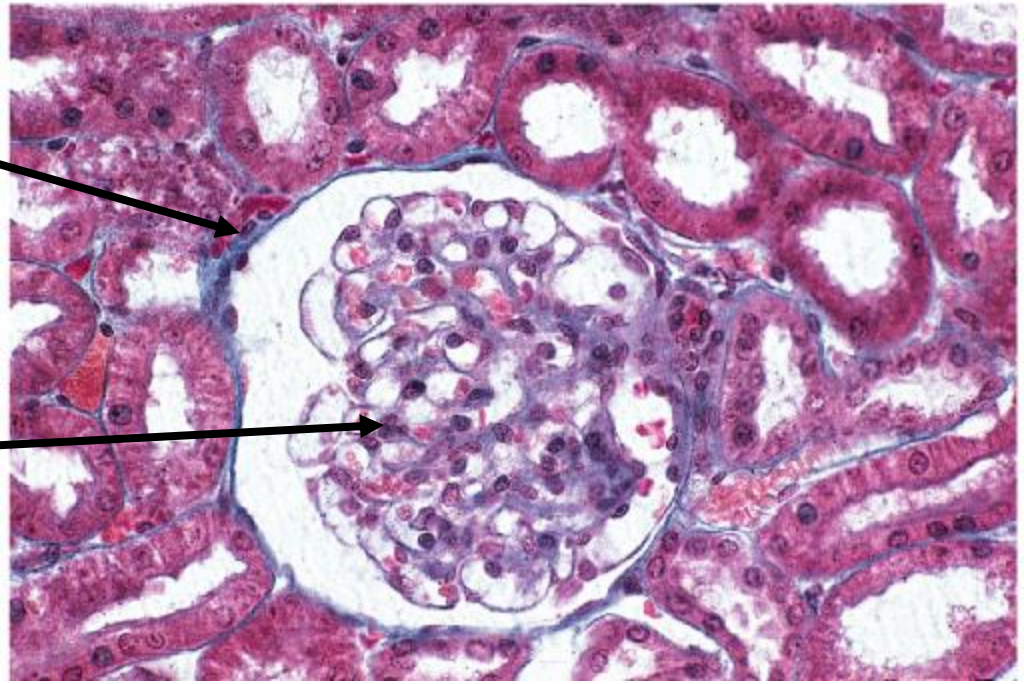
Bowman's capsule



- Outer paietal layer is lined by simple sqamous epithelium
- Inner visceral layer consist of branching epithelial cells - podocytes

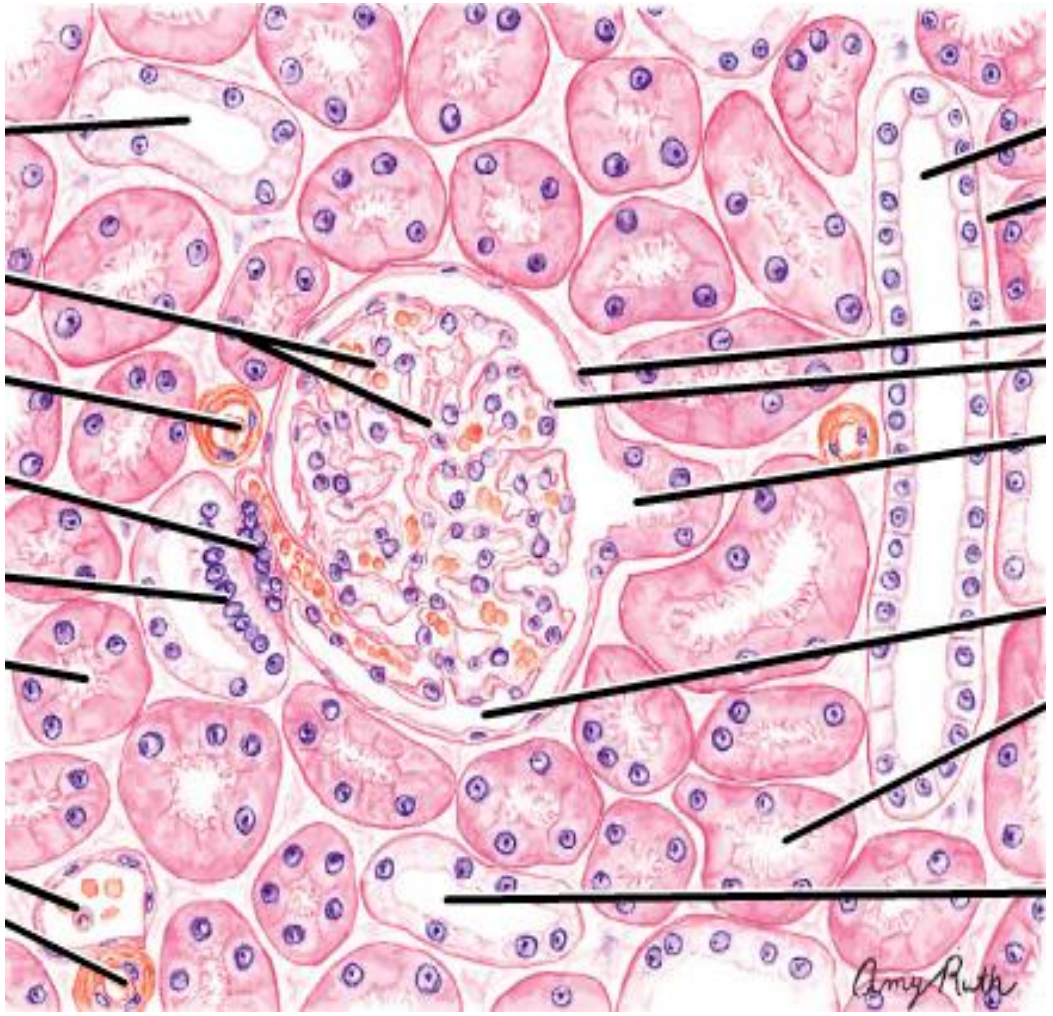
CAPSULE

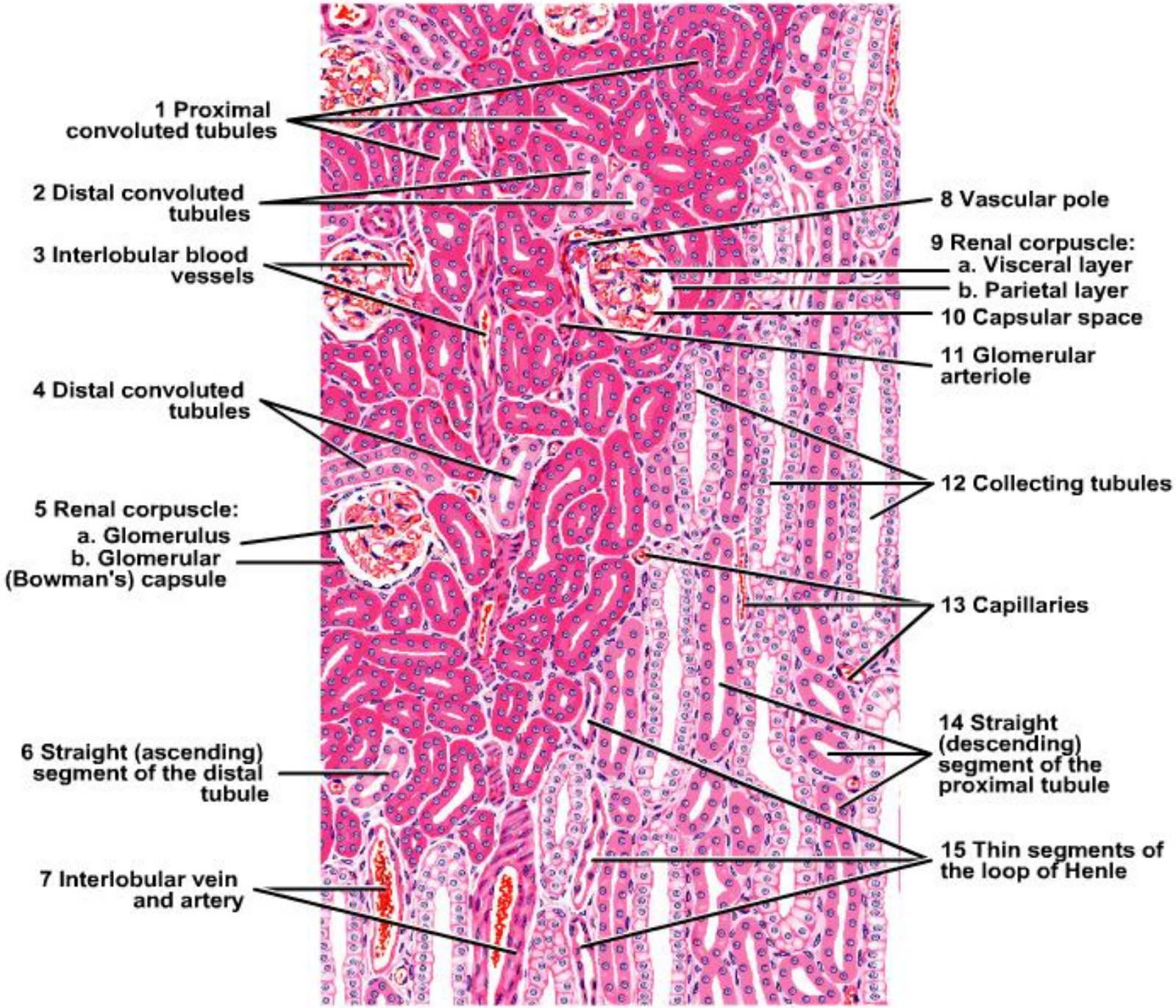
GLOMERULUS



Proximal convoluted tubule:

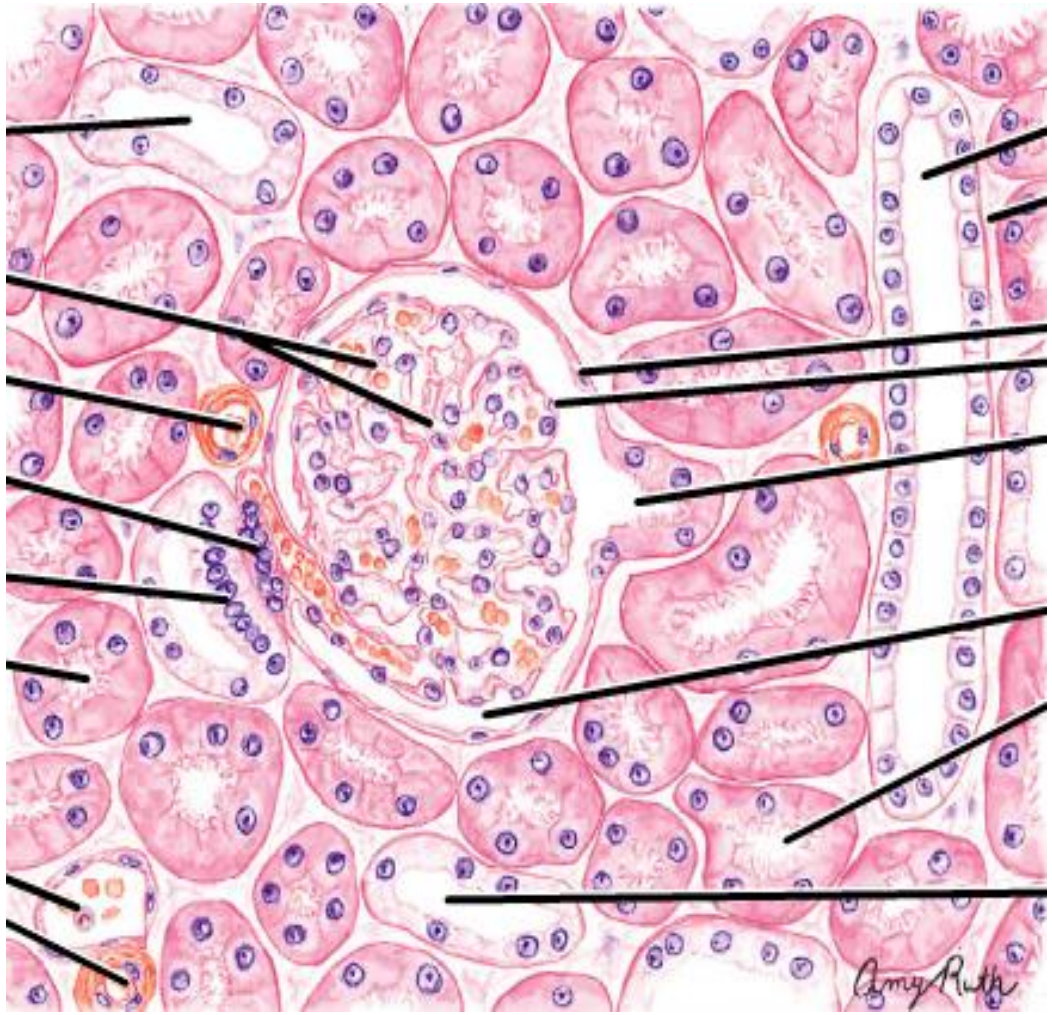
- Lined by simple cuboidal epithelium
- Shows prominent brush border
- Cytoplasm takes intense eosinophilic staining
- Small or uneven lumen

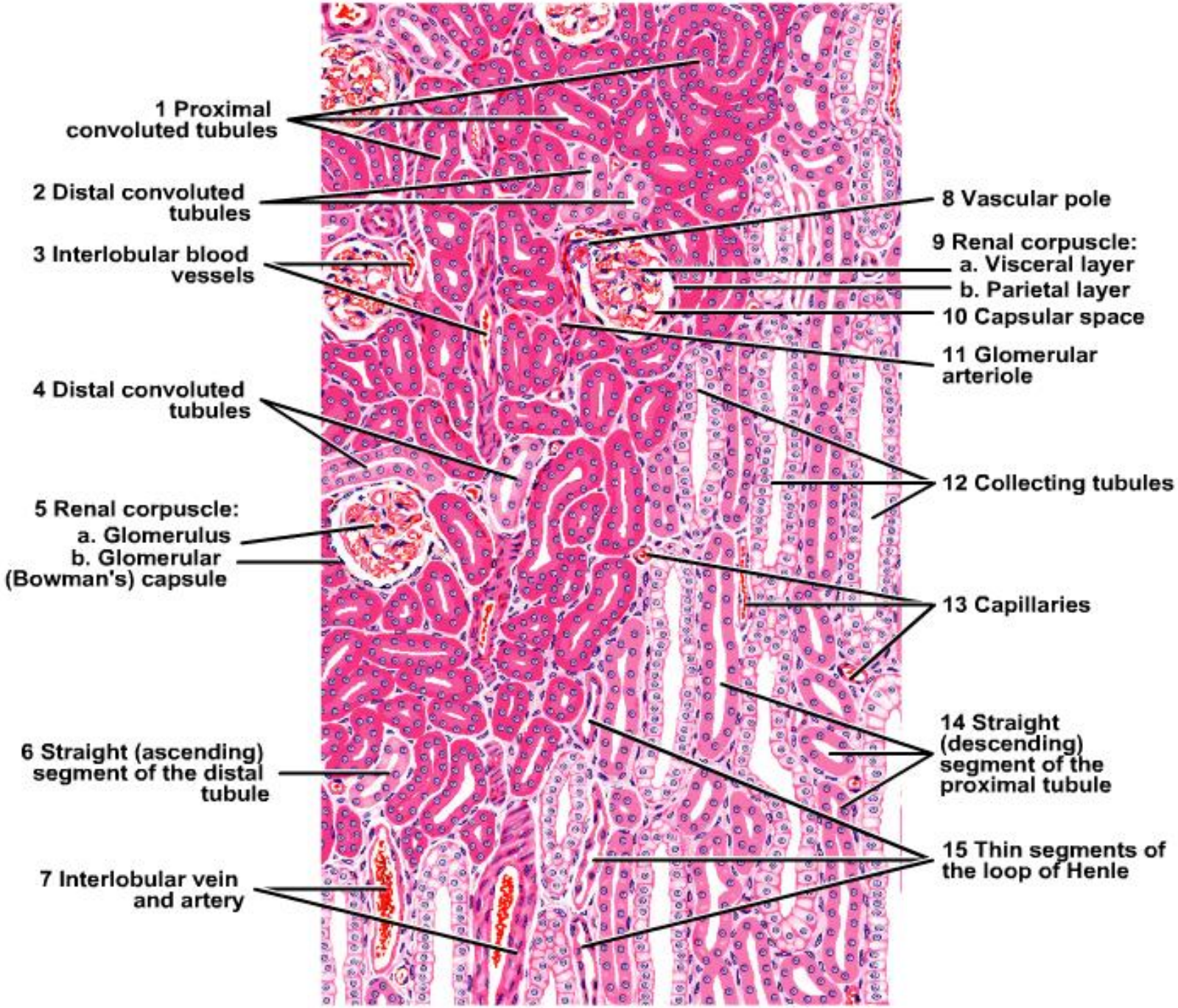



1 Proximal convoluted tubules
2 Distal convoluted tubules
3 Interlobular blood vessels
4 Distal convoluted tubules
5 Renal corpuscle:
a. Glomerulus
b. Glomerular (Bowman's) capsule
6 Straight (ascending) segment of the distal tubule
7 Interlobular vein and artery
8 Vascular pole
9 Renal corpuscle:
a. Visceral layer
b. Parietal layer
10 Capsular space
11 Glomerular arteriole
12 Collecting tubules
13 Capillaries
14 Straight (descending) segment of the proximal tubule
15 Thin segments of the loop of Henle

Distal convoluted tubule:

- Lined by simple cuboidal epithelium
- No brush border
- Clear distinct lumen
- Less intensely stains
- Less in number



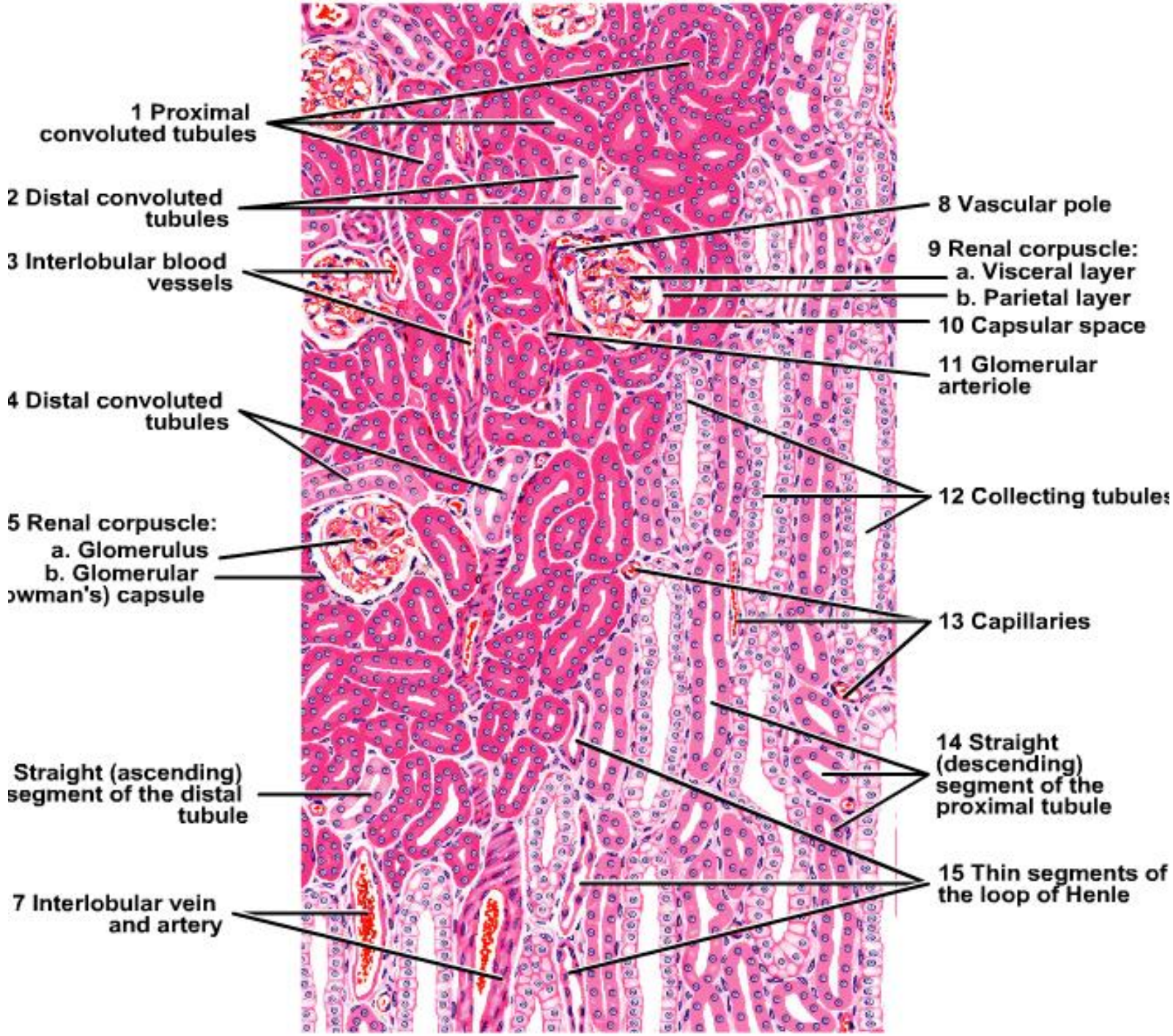

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Collecting tubules:

- Lined by cuboidal epithelium
- Becomes progressively taller
- No brush border

Collecting ducts:

- Formed by the fusion of collecting tubules
- Converge to form larger ducts (ducts of Bellinior papillary ducts)
- Lined by tall columnar epithelium
- Pale stained cytoplasm
- No brush border


1 Proximal convoluted tubules
2 Distal convoluted tubules
3 Interlobular blood vessels
4 Distal convoluted tubules
5 Renal corpuscle:
a. Glomerulus
b. Glomerular (Bowman's) capsule
Straight (ascending) segment of the distal tubule
7 Interlobular vein and artery
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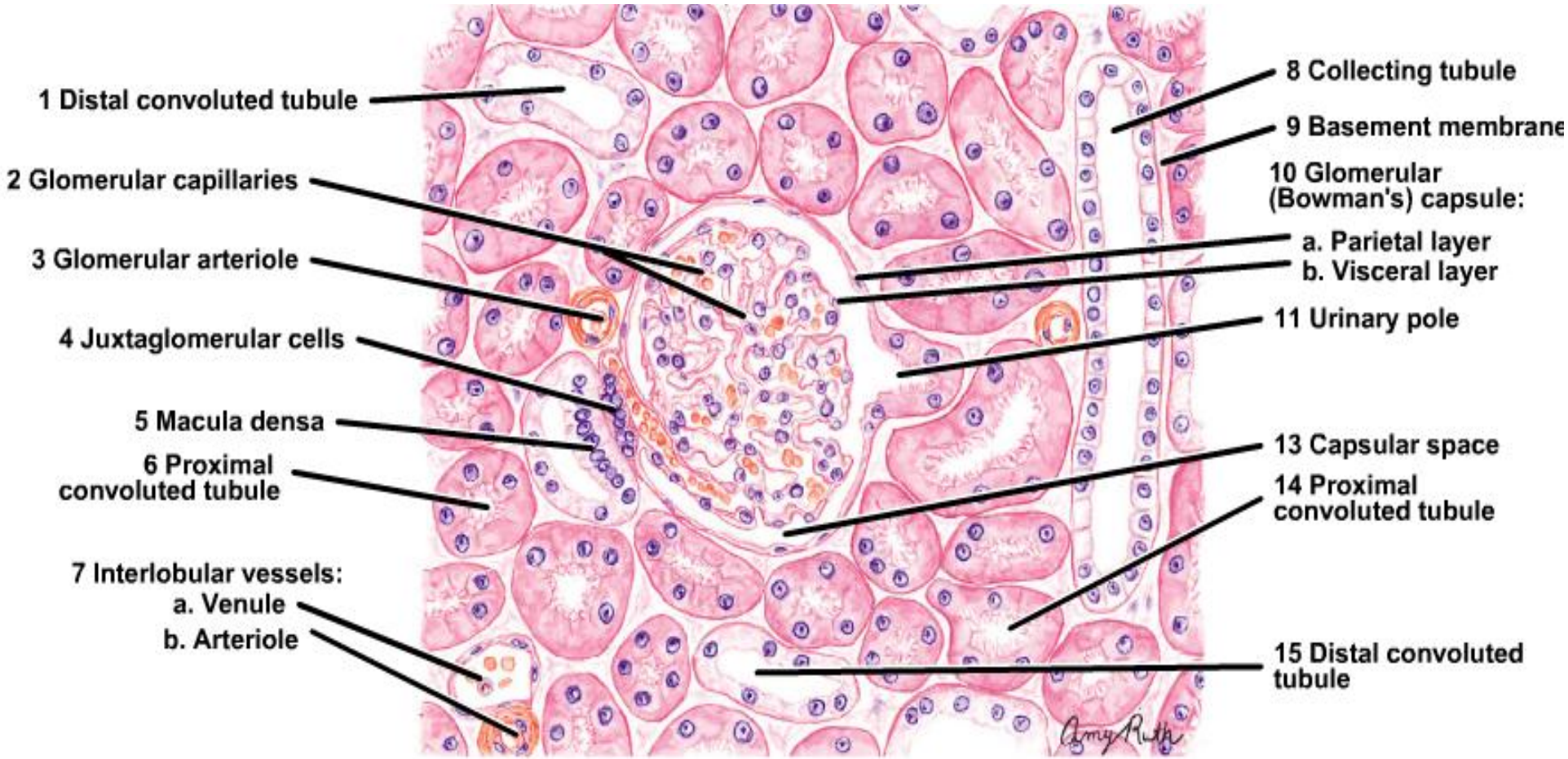
Medulla :

- Contains the segments of the loops of Henle, straight portions of tubules, vasa recta and collecting ducts
- Thin segments of Henle lined by simple squamous epithelium
- Straight portions of tubules are lined by simple cuboidal epithelium
- Collecting ducts (ducts of Bellini/papillary ducts) are lined by simple columnar epithelium

Juxtaglomerular apparatus

- Juxtaglomerular cells: modified Smooth muscle cells in the wall of afferent arteriole
- Macula densa: Modified cells of DCT adjacent to afferent arteriole

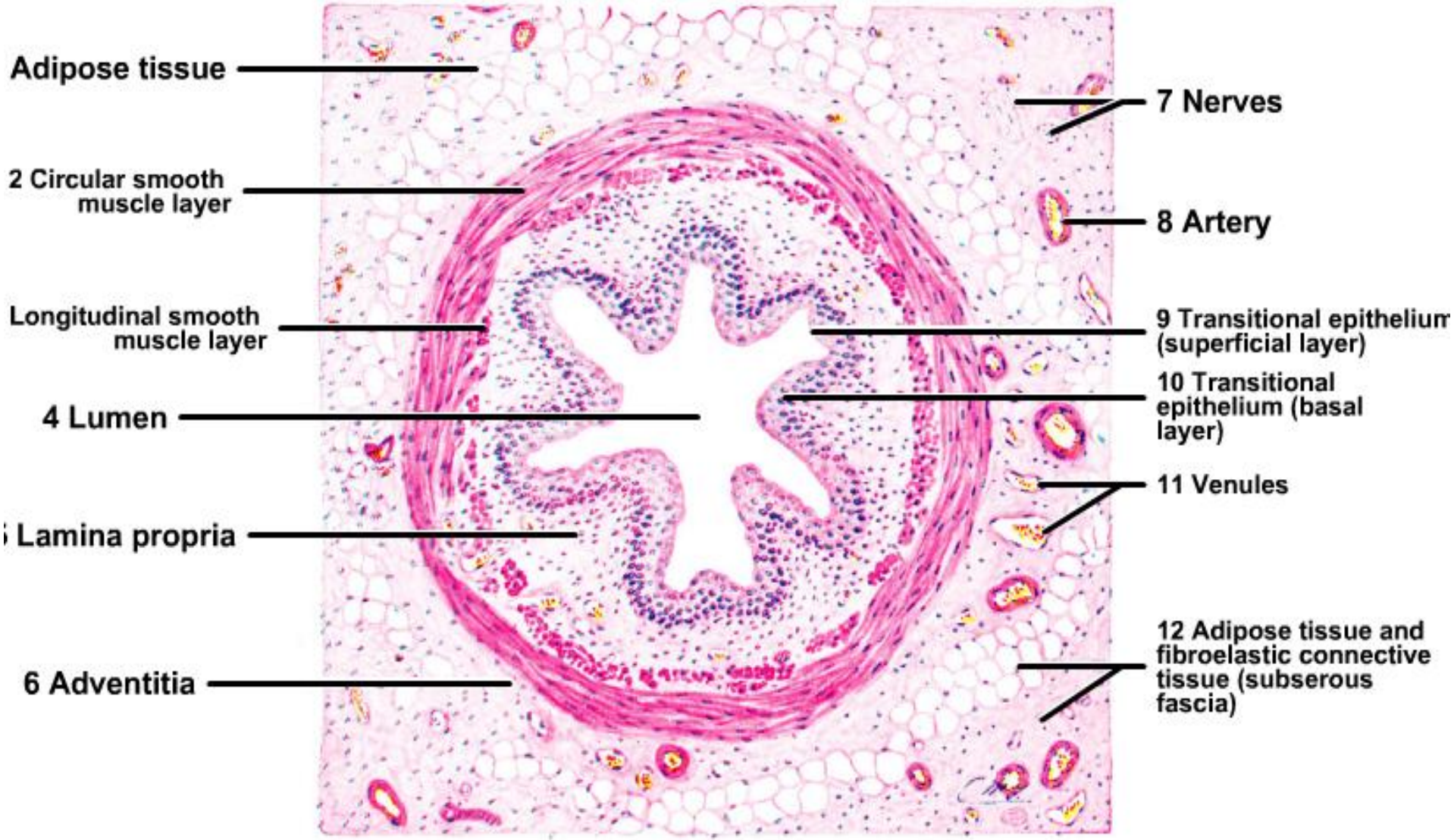



1 Distal convoluted tubule
2 Glomerular capillaries
3 Glomerular arteriole
4 Juxtaglomerular cells
5 Macula densa
6 Proximal convoluted tubule
7 Interlobular vessels:
a. Venule
b. Arteriole
8 Collecting tubule
9 Basement membrane
10 Glomerular (Bowman's) capsule:
a. Parietal layer
b. Visceral layer
11 Urinary pole
13 Capsular space
14 Proximal convoluted tubule
15 Distal convoluted tubule
Amy Ruth

URETER

- Muscular tube
- Mucosa: transitional epithelium
  - Shows numerous folds
  - Wide lamina propria
- Muscularis layer: arranged in two layers – inner longitudinal and outer circular
- Outer coat: Adventitia – loose connective tissue with blood vessels, nerves and lymphatics

URETER


Adipose tissue
2 Circular smooth muscle layer
Longitudinal smooth muscle layer
4 Lumen
Lamina propria
6 Adventitia
7 Nerves
8 Artery
9 Transitional epithelium (superficial layer)
10 Transitional epithelium (basal layer)
11 Venules
12 Adipose tissue and fibroelastic connective tissue (subserous fascia)

URINARY BLADDER

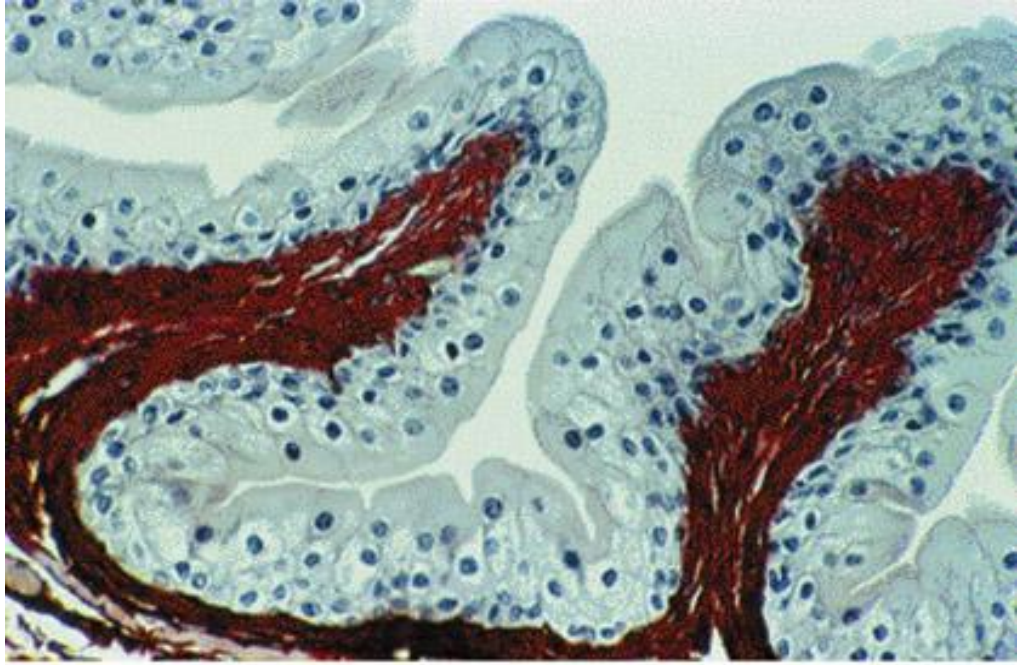
- Mucosa: transitional epithelium

- Muscularis layer:

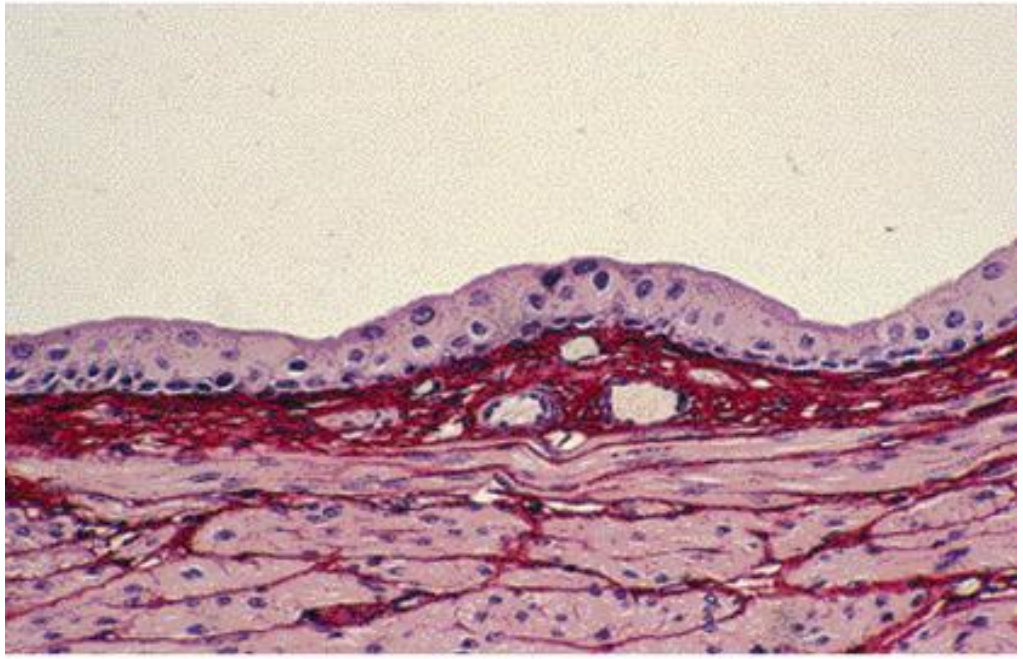
  -Three indistinct layers of smooth muscle and interstitial connective tissue forms meshwork

  -Inner longitudinal, intermediate circular and outer longitudinal

- Outer coat: Serosa or adventitia

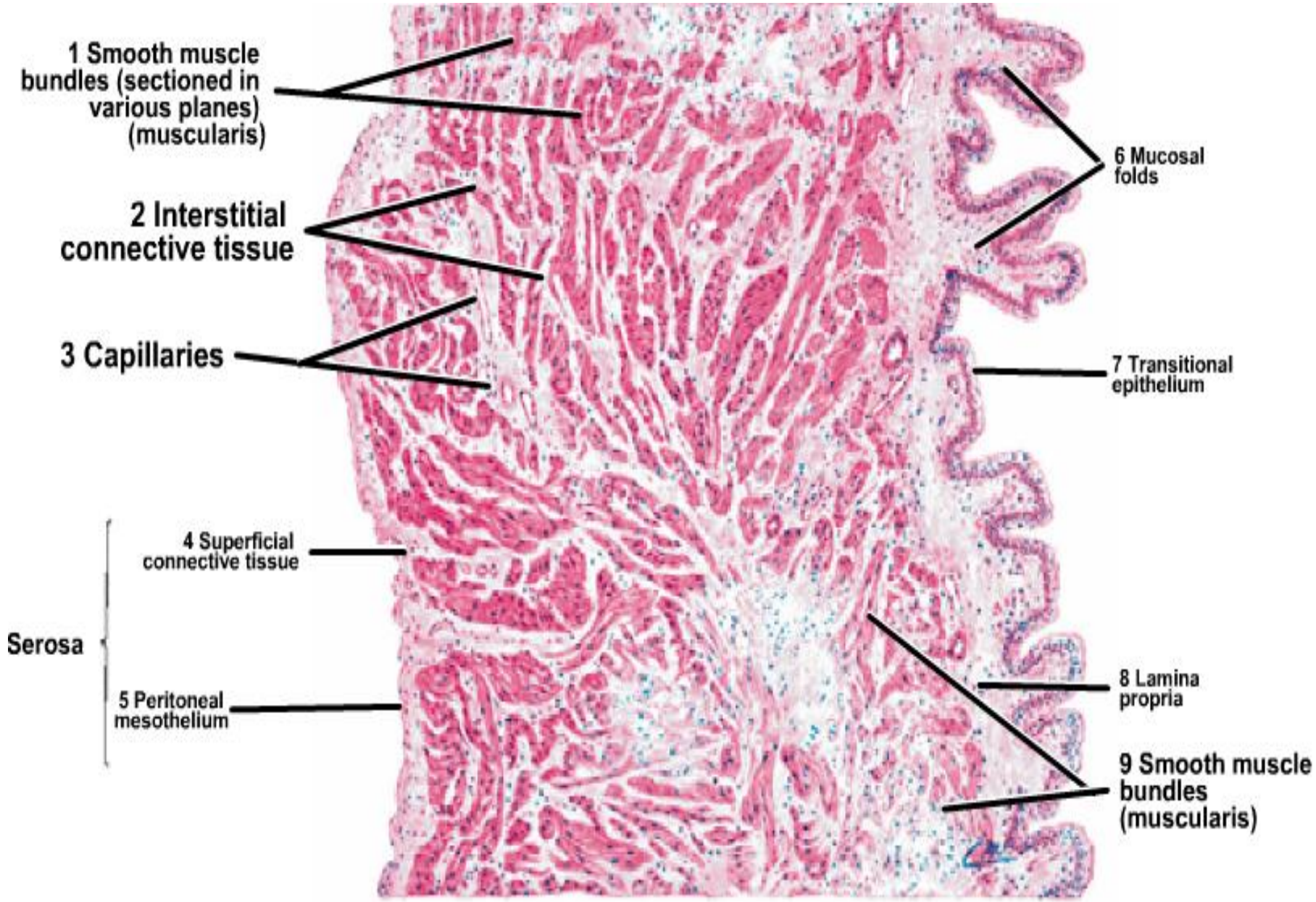


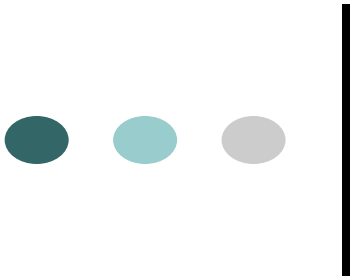
A



B

Transitional epithelium when the urinary bladder is empty (A) or full (B).


1 Smooth muscle bundles (sectioned in various planes) (muscularis)
2 Interstitial connective tissue
3 Capillaries
Serosa
4 Superficial connective tissue
5 Peritoneal mesothelium
6 Mucosal folds
7 Transitional epithelium
8 Lamina propria
9 Smooth muscle bundles (muscularis)



Thank you